

Building an OSS Sandbox using Kuwaiba and OSM



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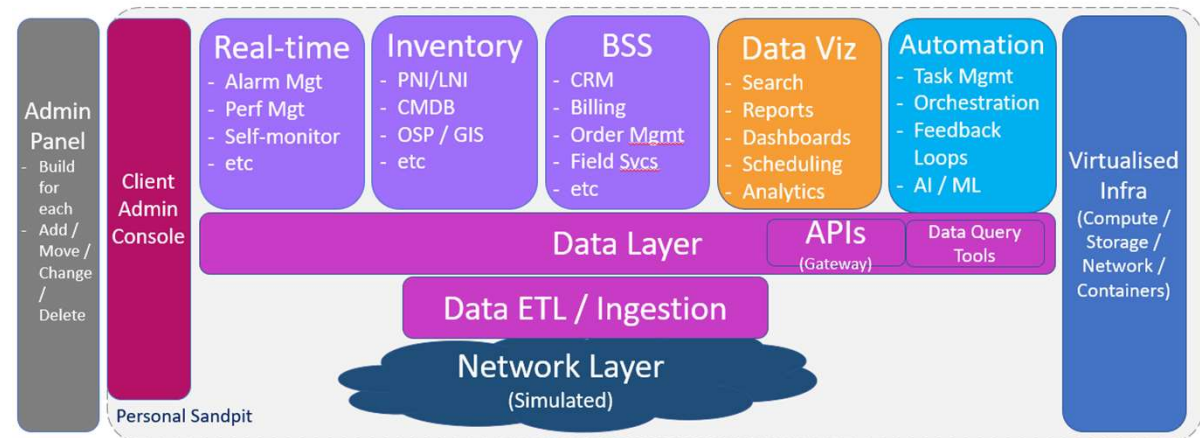


Solent University Networking Lab

Motivation Why do this work?

- We Want to teach realistic Telecoms OSS without major expense at a University
- Want to model real networks and services in a way similar to service providers.
- Want to provide a test environment for Agentic AI and autonomous network prototyping.
- Want any prototype solutions in order to seed real world projects in the Telecoms / ISP space
- Requirements:
 - Easy deploy-ability - docker / cloud
 - Scalability - Large OSS test network
 - Model flexibility - Address Industry Standards
 - Standard API's (ReST / Open API etc)
 - Reference to real world physical networks

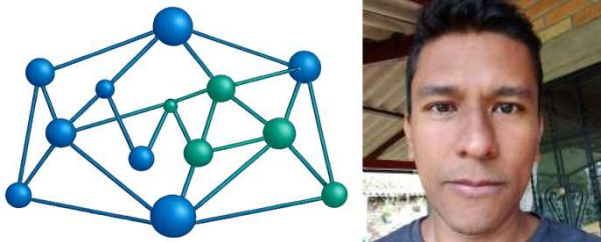
- Inspiration: Passionate About OSS Sandbox (Australia)
- <https://www.linkedin.com/in/ryan-jeffery-oss/>
- <https://passionateaboutoss.com/oss-sandpit-resource-inventory-module/> (Kuwaiba)



Passionate About OSS Sandbox

Kuwaiba Open Source Telecoms Scale Inventory System

- Kuwaiba
 - Developed in Columbia by Neotropic
 - Deployed predominantly with service providers in South America, Asia Pac
 - Architect: Charles Edward Bedón Cortázar (Columbia)
 - <https://www.linkedin.com/in/charles-edward-bed%C3%B3n-cort%C3%A1zar-4525294/>
 - <https://www.neotropic.co/>
 - <https://www.kuwaiba.org/>



- Telecoms inventory System
 - Not just a CMDB
 - Topology and Relationships between elements
 - Asset lifecycle and deployment workflow
 - Very Scalable
 - Comprehensive integration API
 - Neo4J graph database, Docker deployment, Spring boot
 - User Extensible Model

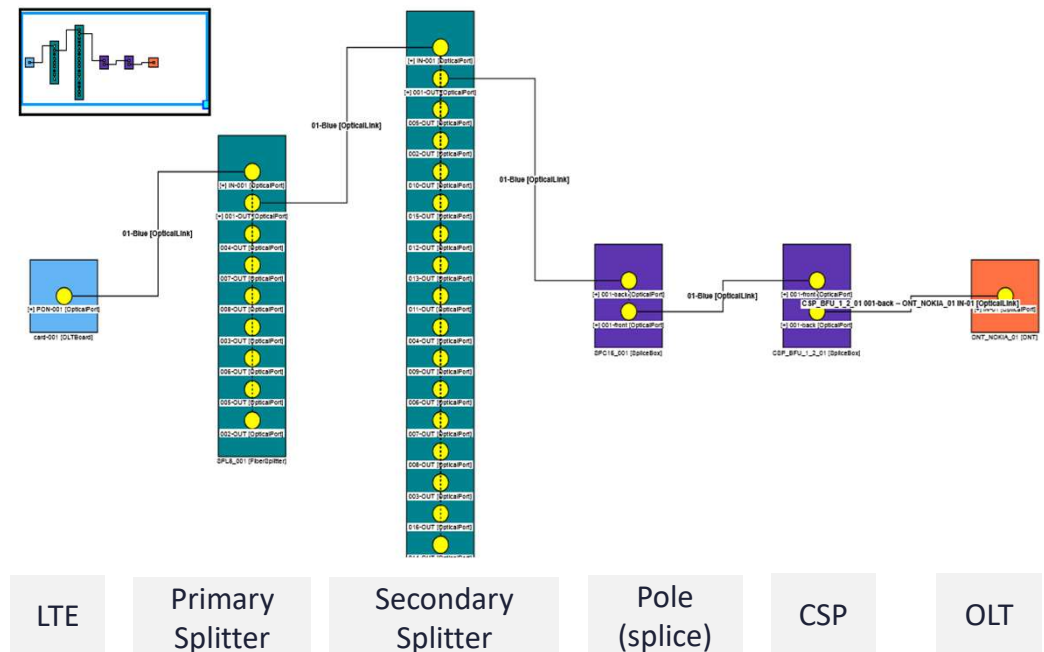
InventoryObject

- CPERouter
- Cloud
- > ■ GenericAppliance
 - MPLSRouter
 - MPLSSwitch
 - Router
 - Switch
- ▼ ■ GenericPhysicalElement
 - BaseBandUnit
 - DWDMmux
 - ExternalEquipment
- > ■ GenericPowerElement
 - OpticalAmplifier
 - OpticalLineTerminal
 - OpticalNetworkTerminal
 - RFModule
 - RadioShelf
- > ■ GenericVirtualElement
- > ■ GenericPart

Kuwaiba Open Source Telecoms Scale Inventory System

- Physical and Logical modelling
 - Templates for Equipment Types, Racks, Cables etc
 - Physical Assets
 - Nodes: Regions, Buildings, Rooms, Racks, Shelves, Connectors, Ports etc
 - Connections: Cable, Fibre, Distribution Frames, Ducts And Pipes
 - Optical, Electrical and Wireless circuits
- Logical assets
 - VLANs, IP addresses, SDH/PDH, MPLS
 - Software components, Cloud, VMs, Containers, Virtual networks
 - Customers, Suppliers, Resource and Customer facing services, Aggregate service offerings

Physical Tree View :: PON-001



Challenge

7 Final Year Students / 3 Months. Sponsor: Passionate about OSS

- **Previous Work: GPON corelation Proof of Concept**
 - GPON Management PoC used OpenNMS with a topology model provided by a UK GPON operator.
 - We could not release the network model which made replication and dissemination difficult.
- **Objectives:**
 - Create a realistic simulated/inferred GPON network topology for a real UK location without using confidential information.
 - Import this into Kuwaiba as the basis for an OSS educator / test bed.
 - Demonstrate the power of Kuwaiba and promote it's use
 - Provide a valuable and realistic student consulting experience addressing a useful problem.
- **GPON Network Modelling**
 - Exchange / POP / Head end
 - Primary / Secondary Splitters
 - Cable Ducting following street geography
 - Cabinets (Splitters / Splices) and Poles not marked but predicted
 - Blown Fibre / Fibre containers Predicted
- **Data Sources**
 - Open Street Map
 - Buildings, Houses, Streets
 - Ordnance Survey
 - UPRN - UK identifier for houses with service

Scope

Usage

- Area: Anywhere with OSM and UPRN data
- Data Sources: OpenStreetMap and Ordnance Survey UPRN

Technology Stack:

- Docker, Spring Boot, Java, PostGIS, pgRouting

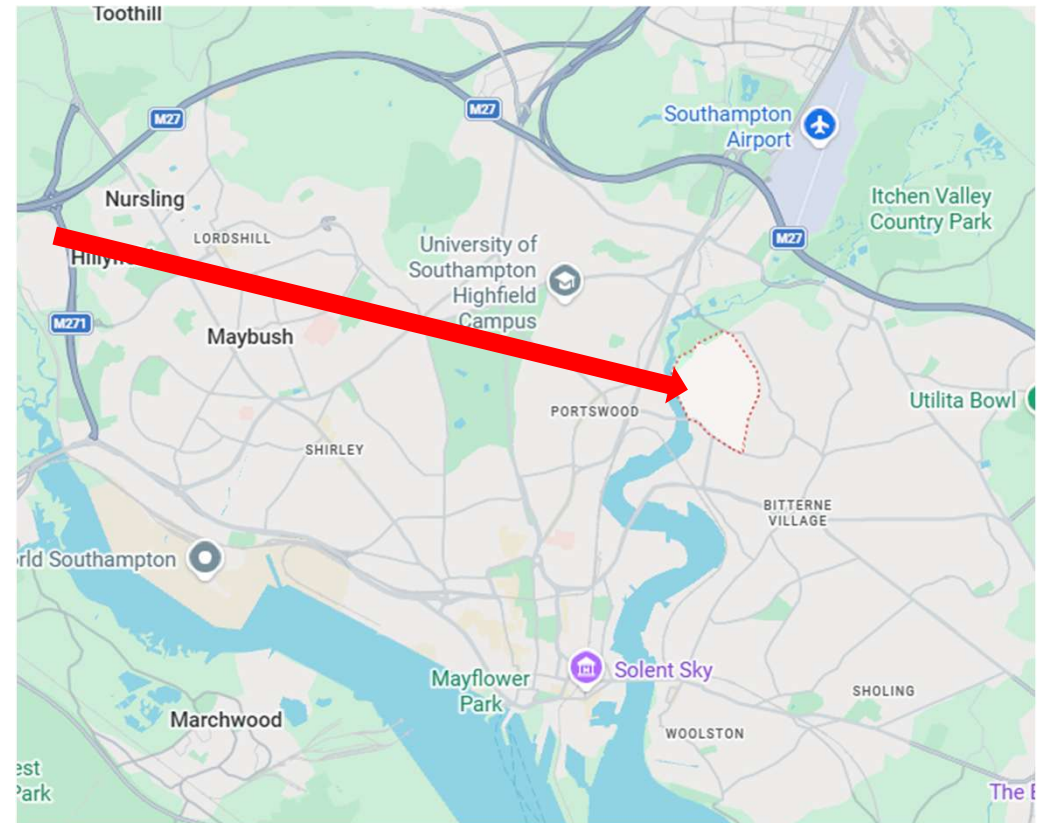
Network Hierarchy:

- Exchange, Aggregator, Cabinet, Pole, Building

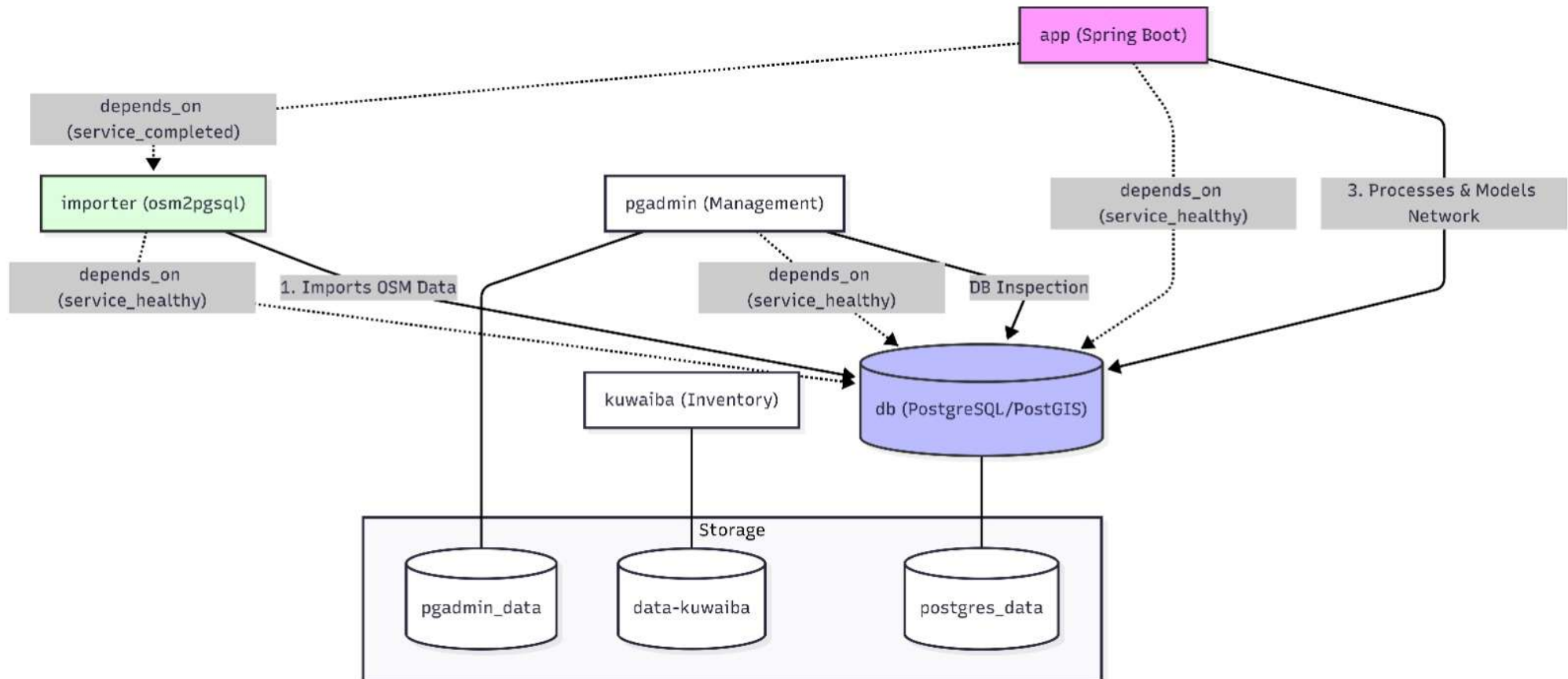
Exclusions

- Static data
- Terrain height
- Existing Infrastructure

Bitterne Park Southampton



System Architecture Part 1 - Docker Stack



System Architecture

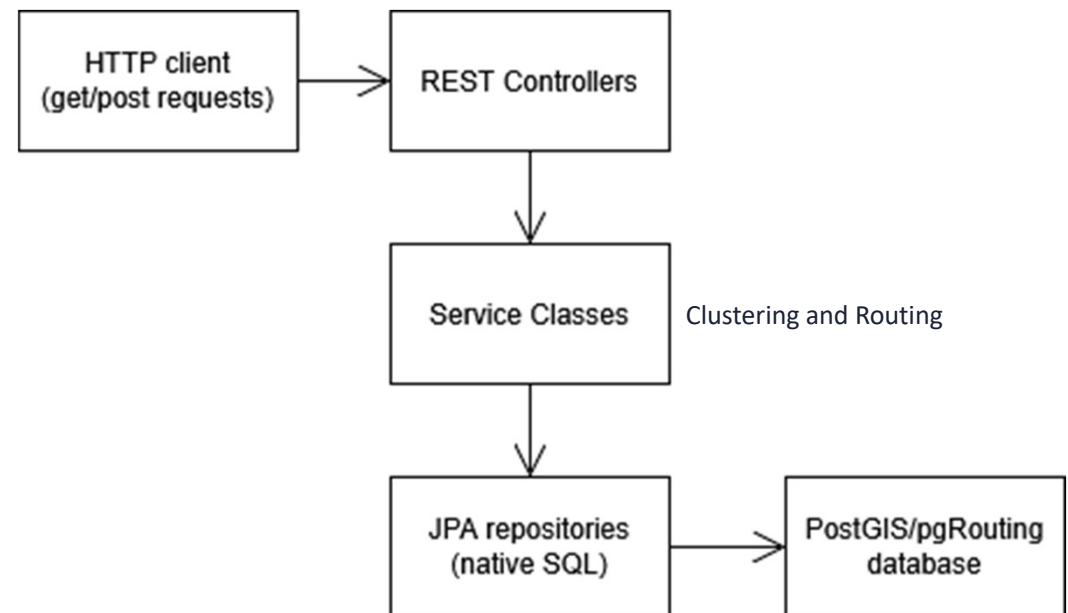
Part 2 - Spring Boot App

Layered Pattern Architecture

Schema Initialization on startup with SQL to
pgRouting topology

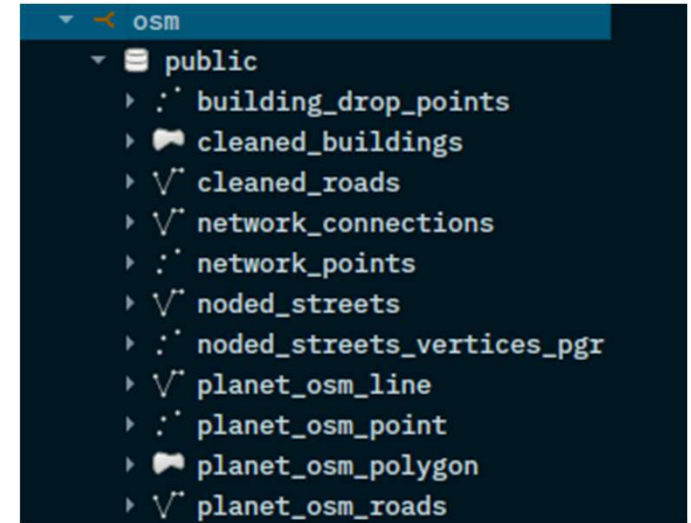
Custom Jackson Serializers

(JTS geometry -> GeoJSON)



Data Processing & Cleaning

- Raw OpenStreetMap imported into database
- Relevant data moved into usable tables
- New tables have spatial indexing
- Ready for infrastructure prediction
- Identified bottleneck regarding street name data enrichment.



Infrastructure Prediction

Phase 1 - Bottom Up Clustering

- First Phase: Bottom-up clustering
 - PostGIS K-Means Clustering

Infrastructure Level	Input Points	Ratio
Pole	Building Drop Points	12 buildings per pole
Cabinet	Poles	8 poles per cabinet
Aggregator	Cabinets	8 cabinets per aggregator
Exchange	Aggregators	8 aggregators per exchange

Infrastructure Prediction

Phase 2 Top-down Connection Routing

- Dijkstra routing algorithm used
- If routing fails, there is a straight-line fallback connection
- DROP connections are straight lines mimicking real life pole to building connections

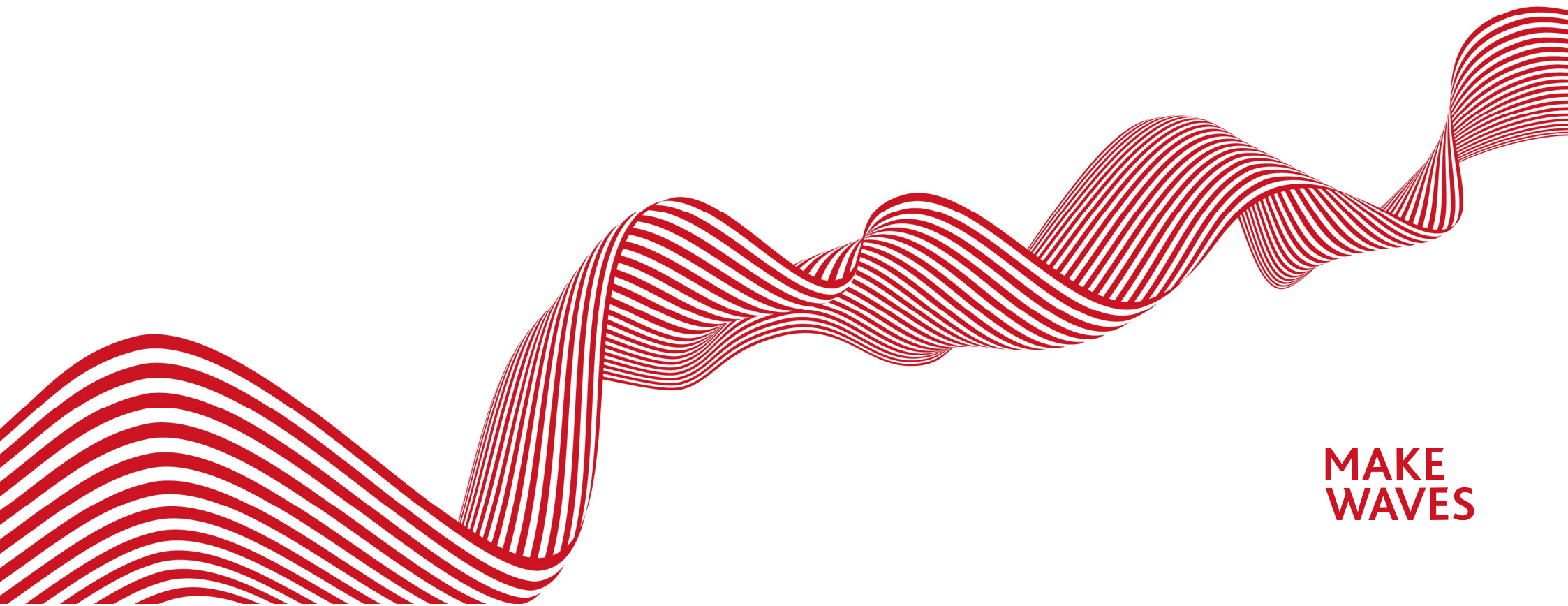
Connection Type	From	To	Geometry Method
DROP	Pole	Building Drop Point	ST_MakeLine(direct)
FEEDER	Cabinet	Pole	pgr_dijkstra (road-following)
DISTRIBUTION	Aggregator	Cabinet	pgr_dijkstra (road-following)
TRUNK	Exchange	Aggregator	pgr_dijkstra (road-following)

QGIS View of predicted infrastructure



Generated Ducts, Cabinets, Poles and Drop Points to homes

Demo...



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Next Steps

The bigger picture - next generation cloud native OSS

1. Tidy up the current GPON network importer

- Make automated network design more configurable
- allow users to modify ratios of splitters to end points.
- Support flexible naming conventions for the generated network

2. Extend the OSS Sandbox concept

- Integrate a Cloud Native Monitoring Layer
- Delta-v - (OpenNMS as Microservices)
- Support service providers migrating to CNCF / Telecoms Infrastructure Project (TIP) architectures
- OpenRAN / TIP cloud native OSS
- Support streaming telemetry and flows-based traffic management
- Excellent legacy SNMP support

3. Future Projects

- GPON fault corelation using MSP / AI integration
- TM Forum Open Digital Architecture
- ODA canvas Kubernetes components
- Network as a Service API's
- Resource / Service Inventory (Kuwaiba)
- Fault Management, Service Problem Management (Delta-v)

References

Students Github Project: <https://github.com/com617-industrial-2025-1/com617-kuwaiba-inventory-1>

UPRN <https://www.ordnancesurvey.co.uk/public/unique-property-reference-numbers>

CARLINO, D., *et al.*, *osm2streets* Available from: <https://github.com/a-b-street/osm2streets>

OSM2PGSQL, 2026. *osm2pgsql* [viewed May 7 2026]. Available from: <https://osm2pgsql.org/>

TM Forum Open Digital Architecture <https://www.tmforum.org/open-digital-architecture/>

TM Forum Network as a Service NaaS Requirements <https://www.tmforum.org/resources/standard/ig1169-network-as-a-service-naas-requirements-r18-0-0/>

OpenNMS as Microservices - Delta-V <https://github.com/pbrane/delta-v>

Kuwaiba <https://www.kuwaiba.org/>

Kuwaiba Code: <https://sourceforge.net/projects/kuwaiba/>

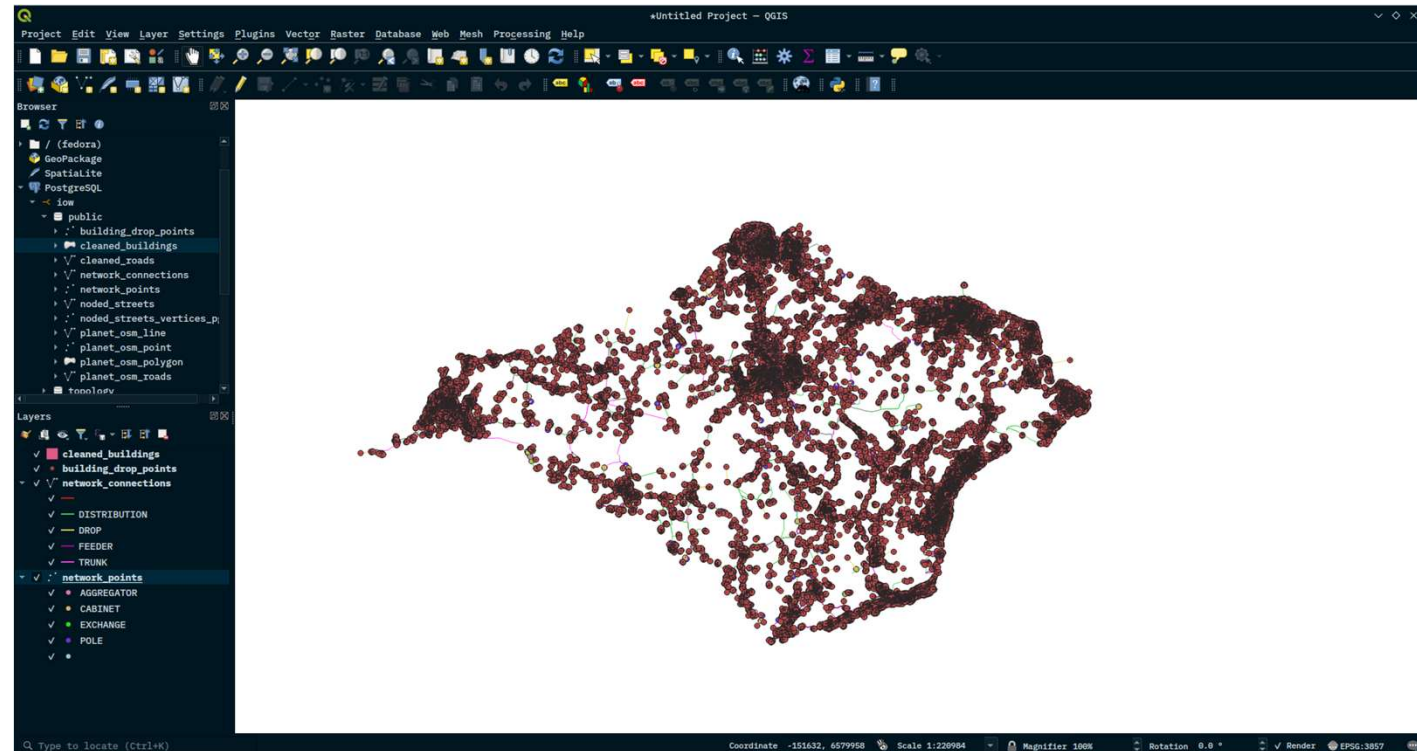
Thank you for listening



Backup Slides



Scalability



Dataset	Source	Buildings	Pipeline Time
Winchester	Osm2streets	~500	3s
Isle of Wight	Geofabrik	~50,000	16m 39s
Hampshire	Geofabrik	614,711	Did not complete